The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 July 2006.

INCH-POUND

MIL-PRF-19500/616D
3 April 2006
SUPERSEDING
MIL-PRF-19500/616C
21 March 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, DUAL, COMMON CATHODE OR ANODE CENTER TAP, ULTRAFAST, TYPES 1N6657 THROUGH 1N6659 AND 1N6657R THROUGH 1N6659R, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, dual high voltage, ultrafast power rectifier diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-254AA isolated).

1.3 Maximum ratings. (per leg)

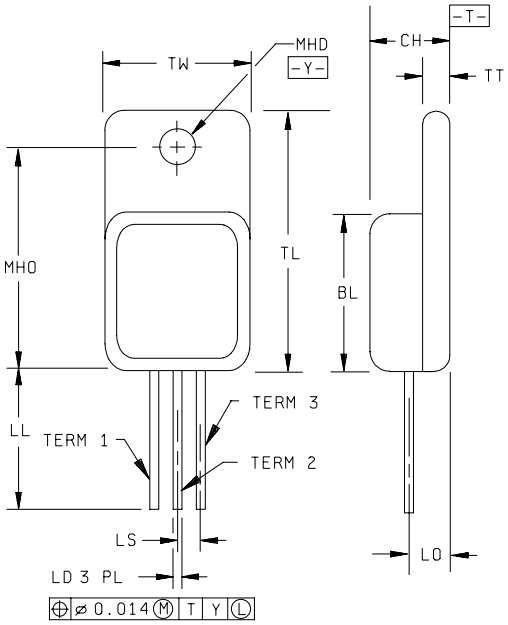
| Types | $V_{RWM}$ | $I_{FSM}$ (1) $t_p$ = 8.3 ms | $I_F$ $T_C$ = 100°C (1) (2) (3) | $t_{rr}$ (1) | $R_{\theta JC}$ (1) | $R_{\theta JA}$ (1) | $T_{STG}$ and $T_J$ |
|---|---|---|---|---|---|---|---|
| | V dc | A (pk) | A dc | ns | °C/W | °C/W | °C |
| 1N6657, 1N6657R | 100 | | | | | | |
| 1N6658, 1N6658R | 150 | 150 | 15 | 35 | 2.3 | 40 | -65 to +200 |
| 1N6659, 1N6659R | 200 | | | | | | |

(1) Each individual diode.
(2) Derate linearly at 300 mA/°C from +100°C to +150°C.
(3) Total package current is limited to 30A dc.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.



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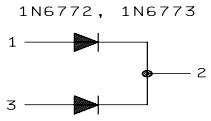



| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .535 | .545 | 13.59 | 13.84 |
| CH | .249 | .260 | 6.32 | 6.60 |
| LD | .035 | .045 | 0.89 | 1.14 |
| LL | .530 | .550 | 13.46 | 13.97 |
| LO | .150 BSC | | 3.81 BSC | |
| LS | .150 BSC | | 3.81 BSC | |
| MHD | .139 | .149 | 3.53 | 3.78 |
| MHO | .665 | .685 | 16.89 | 17.40 |
| TL | .790 | .800 | 20.07 | 20.32 |
| TT | .040 | .050 | 1.02 | 1.27 |
| TW | .535 | .545 | 13.59 | 13.84 |

SCHEMATIC

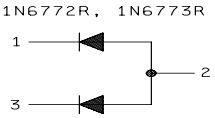
1N6657, 1N6658, 1N6659

TERM 1 = ANODE 1
TERM 2 = CATHODE
TERM 3 = ANODE 2




1N6657R, 1N6658R, 1N6659R

TERM 1 = CATHODE 1
TERM 2 = ANODE
TERM 3 = CATHODE 2




NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. All terminals are isolated from case.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Dimensions and configuration (TO-254AA).



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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil. or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-254AA) herein. Methods used for electrical isolation of the terminal feed throughs shall employ materials that contain a minimum of 90 percent $AL_2O_3$ (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages. The US Government's preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.



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3.4.1 Lead formation and finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of MIL-PRF-19500.

3.4.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.2.1.1 Group E thermal impedance. Each supplier shall submit a thermal impedance ($Z_{\theta JX}$) histogram of the entire qualification lot. The histogram data shall be taken prior to the removal of devices that are atypical for thermal impedance. Thermal impedance curves (from $Z_{\theta JX}$ test pulse time to $R_{\theta JX}$ minimum steady-state time) of the best device in the qual lot and the worst device in the qual lot (that meets the supplier proposed screening limit), or from the thermal grouping, shall be submitted. The optimal test conditions and proposed initial thermal impedance screening limit shall be provided in the qualification report. Data indicating how the optimal test conditions were derived for $Z_{\theta JX}$ shall also be submitted. The proposed specification maximum thermal impedance curve shall be submitted. The qualifying activity may approve a different $Z_{\theta JX}$ limit not to exceed the specification's thermal curve for conformance inspection end-point measurements as applicable**.** Equivalent data, procedures, or statistical process control plans may be used for part, or all, of the above requirements. The approved thermal impedance conditions and limit for $Z_{\theta JX}$ shall be used by the supplier in screening and table I, subgroup 2. The approved thermal resistance conditions for $R_{\theta JX}$ shall be used by the supplier for conformance inspection. For product families with similar thermal characteristics based on the same physical and thermal die, package, and construction combination (thermal grouping), the supplier may use the same thermal impedance curves.



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* 4.3 Screening ( JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with appendix E, table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see appendix E, table IV of MIL-PRF-19500 | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 1a<br>1b | Required<br>Required | Not required<br>Required (JANTXV only) |
| 2 | Not required | Not required |
| 3a<br>3b<br>(1) 3c | Required<br>Surge (see 4.3.1)<br>Thermal impedance (see 4.3.2) | Required<br>Surge (see 4.3.1)<br>Thermal impedance (see 4.3.2) |
| 4, 5, 6 | Not applicable | Not applicable |
| 7a<br>7b | Optional<br>Optional | Optional<br>Optional |
| 8 | Required | Not applicable |
| 9 | $V_{F1}$ and $I_{R1}$ | Not applicable |
| 10 | Method 1038 of MIL-STD-750, test condition A; t = 48 hours; $V_R$ = 80 percent of rated $V_R$. | Not applicable |
| 11 | $V_{F1}$ and $I_{R1;}$ $\Delta V_{F1}$ = ± 0.1 V (pk); $\Delta I_{R1}$ = ± 2 µA dc or 100 percent from the initial value; whichever is greater. | $V_{F1}$ and $I_{R1}$ |
| 12 | Method 1038 of MIL-STD-750, test condition B; t = 240 hours; (see 4.3.3) | Method 1038 of MIL-STD-750, test condition A; t = 48 hours; $V_R$ = 80 percent of rated $V_R$. |
| 13 | Subgroup 2 and 3 of table I herein; $V_{F1}$ and $I_{R1}$; $\Delta V_{F1}$ = ± 0.1 V (pk); $\Delta I_{R1}$ = ± 2 µA dc or 100 percent from the initial value; whichever is greater. | Subgroup 2 of table I herein; $V_{F1}$ and $I_{R1}$; $\Delta V_{F1}$ = ± 0.1 V (pk); $\Delta I_{R1}$ = ± 2 µA dc or 100 percent from the initial value; whichever is greater. |
| 14a<br>14b | Required<br>Required | Required<br>Required |
| 15 | Required | Not required |
| 16 | Required | Not required |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Surge current. Surge current, method 4066 of MIL-STD-750. $I_O$ = 0; $V_{RM}(w)$ = 0; $I_{FSM}$ = see 1.3; six surges; $T_A$ = 25°C, $t_p$ = 8.3 ms, one minute minimum time between surges.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081, as applicable, of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ ($V_C$ and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 µs max. See table II, group E, subgroup 4 herein.

* 4.3.3 Burn-in conditions. Burn-in conditions are as follows: $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. $V_R$ = rated. Adjust $I_O$ to achieve $T_J$ =150°C minimum.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.



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* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions of appendix E, tables VIa (JANS) and VIb (JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (endpoints) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS of MIL-PRF-19500).

| Subgroup | Method | Condition |
|---|---|---|
| B4 | 1037 | $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. $I_F$ or $I_O$ = 1.25 to 10 A. Minimum for 2,000 cycles. |

4.4.2.2 Group B inspection, appendix E, table VIb (JANTX and JANTXV of MIL-PRF-19500).

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1037 | $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. $I_F$ or $I_O$ = 1.25 to 10 A. Minimum for 2,000 cycles. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (endpoints) shall be in accordance with table I, subgroup 2 herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition A, weight = 10 pounds, t = 15 seconds. |
| C5 | 3101<br>or<br>4081 | $R_{\theta JC}$ = 2.3°C/W. |
| C6 | 1037 | $T_A$ = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. $I_F$ or $I_O$ = 1.25 to 10 A for 6,000 cycles. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and table II herein. Electrical measurements (endpoints) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Burn-in and steady-state operation life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectifier current. The forward conduction angle of the rectified current shall not be greater than 180 degrees nor less than 150 degrees.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3101 | See 4.3.2 | $Z_{\theta JX}$ | | | °C/W |
| Breakdown voltage | 4022 | $I_R$ = 500 μA dc; pulsed (see 4.5.1) | $V_{BR}$ | | | V dc |
| 1N6657, 1N6657R<br>1N6658, 1N6658R<br>1N6659, 1N6659R | | | | 100<br>150<br>200 | | |
| Forward voltage | 4011 | $I_F$ = 10 A (pk); pulsed (see 4.5.1) | $V_{F1}$ | | 1.0 | V dc |
| | | $I_F$ = 20 A (pk); pulsed (see 4.5.1) | $V_{F2}$ | | 1.2 | V dc |
| Reverse current leakage | 4016 | DC method; $V_R$ = rated $V_R$, (see 1.3); pulsed (see 4.5.1) | $I_{R1}$ | | 10 | μA dc |
| Subgroup 3 | | | | | | |
| High temperature operation | | $T_C$ = +100°C | | | | |
| Reverse current leakage | 4016 | DC method; $V_R$ = rated $V_R$ (see 1.3); pulsed (see 4.5.1) | $I_{R2}$ | | 1.0 | mA dc |
| Low temperature operation: | | $T_A$ = -65°C | | | | |
| Forward voltage | 4011 | $I_F$ = 10 A (pk); pulsed (see 4.5.1) | $V_{F3}$ | | 1.15 | V dc |
| Subgroup 4 | | | | | | |
| Scope display evaluation 2/ | | | | | | |
| Reverse recovery time | 4031 | Condition B; $I_F$ = 1A dc, $I_R$ = 1 A, $I_{RR}$ = 100 mA | $t_{rr}$ | | 35 | ns |
| Junction capacitance | 4001 | $V_R$ = 10 V dc, f = 1 mHz, $V_{SIG}$ = 50 mV (p-p) (max) | $C_J$ | | 150 | |
| Subgroup 5 | | | | | | |
| Not applicable | | | | | | |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 6<br>Surge | 4066 | Mounting conditions in accordance with test method 1026 of MIL-STD-750, $T_A$ = +25°C, $I_{FSM}$ = (see 1.3), $I_O$ = 0; $V_{RM}(w)$ = 0; six surges; $T_A$ = 25°C, $t_p$ = 8.3 ms, one minute minimum time between surges. | | | | |
| Subgroup 7<br>Not applicable | | | | | | |

1/ For sampling plan, see MIL-PRF-19500.
2/ The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 μA/division and 20 to 50 V/division. Reverse current over the knee shall be at least 500 μA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.



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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 22 devices<br>c = 0 |
| Temperature cycling | 1051 | 500 cycles, condition C, - 55°C to +175°C. | |
| Hermetic seal | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |
| Subgroup 2 | | | 22 devices<br>c = 0 |
| Blocking life | 1048 | 1,000 hours, $V_R$ = 80 percent of $V_{Rated}$. | |
| Electrical measurements | | See table I, subgroup 2 herein except for thermal impedance. | |
| Subgroup 4 | | | Sample size<br>N/A |
| Thermal impedance curves | | See 4.2.1.1 | |
| Subgroup 5 | | | 22 devices<br>c = 0 |
| Barometric pressure, reduced (altitude operation) | 1001 | $V_R$ = rated $V_R$ (see 1.3); pressure = 33 mmHg; t = 1 minute (minimum), $R_{ISO}$ = 2.0 x $10^6$ ohm max. | |
| Subgroup 6 | | | |
| Not applicable | | | |
| Subgroup 8 | | | 22 devices<br>c = 0 |
| Forward surge | 4066 | | |
| Electrical measurements | | See table I, subgroup 2 herein. | |



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Interchangeability information. MIL-PRF-19500/616 is a dual TO-254 package version of MIL-PRF-19500/478, which is a stud package version.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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